

Geetha Bommareddy

geethabommareddy99@gmail.com | +1 (470)923-3125 | [LinkedIn](#) | [Portfolio](#)

SUMMARY

AI/ML Engineer with 4+ years of experience building and deploying scalable machine learning and generative AI solutions in the financial services domain. Expertise in real-time fraud detection, risk modeling, and large-scale data pipelines, processing over 100K+ transactions daily with low-latency inference systems. Strong in Generative AI and LLM engineering, including Retrieval-Augmented Generation (RAG), semantic search, and knowledge assistants using GPT and Hugging Face. Proven ability to design production-grade ML systems with MLOps, Kubernetes, and cloud platforms (AWS, Databricks), delivering measurable impact such as reducing inference time by 60% and detecting 8,000+ anomalies per quarter. Passionate about building intelligent, scalable, and explainable AI systems in regulated environments.

SKILLS

Artificial Intelligence & Machine Learning: Supervised & Unsupervised Learning, Ensemble Methods (XGBoost, LightGBM, CatBoost), Anomaly Detection, Time Series Forecasting (Prophet, ARIMA, LSTM), Feature Engineering, Model Optimization, Explainable AI

Generative AI & LLM Engineering: Large Language Models (GPT, BERT, LLaMA), Retrieval-Augmented Generation (RAG), Agentic AI Workflows, Prompt Engineering, Context Optimization, Embedding Models, Semantic Search, Vector Databases (Pinecone, FAISS, Weaviate), LangChain, LlamaIndex, Knowledge Retrieval Systems

Deep Learning & Advanced AI: Transformer Architectures, CNNs, RNNs, LSTMs, Sequence Modeling, Autoencoders, NLP (NER, Text Classification, Summarization), PyTorch Lightning, TensorFlow, Hugging Face Transformers

MLOps & AI Production Systems: End-to-End ML Pipelines, Model Deployment & Monitoring, MLflow, CI/CD (GitHub Actions, Jenkins), Docker, Kubernetes (EKS), Feature Stores, Model Versioning, Drift Detection, A/B Testing, AI Observability

Data Engineering & Real-Time Processing: PySpark, Apache Spark, Kafka (Streaming Pipelines), Airflow (Workflow Orchestration), ETL/ELT Pipelines, Data Modeling, Data Quality & Validation, Batch & Real-Time Processing

Cloud & Data Platforms: AWS (SageMaker, S3, Glue, Redshift), Azure Machine Learning, Databricks, Snowflake, Distributed Computing

Programming & Frameworks: Python, SQL, Pandas, NumPy, Scikit-learn, FastAPI, Flask, REST APIs, JupyterLab, Git

Data Visualization & Analytics: Tableau, Power BI, Plotly, Dash, Matplotlib, Advanced Excel

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB

Certifications: [AWS CERTIFIED DATA ENGINEER](#)

EXPERIENCE

JP Morgan Chase & Co. | AI/ML Engineer

Jan 2024 – Present | USA

- Designed and deployed real-time risk scoring and fraud detection systems using Python, XGBoost, and PySpark, enabling sub-3-second inference for high-volume transaction streams (100K+ daily events).
- Built streaming ML pipelines with Kafka and Spark Structured Streaming, performing real-time feature engineering and anomaly detection to support 24x7 fraud monitoring operations.
- Developed LLM-powered knowledge assistants using GPT, Hugging Face Transformers, and Retrieval-Augmented Generation (RAG), enabling semantic search and automated summarization of policies, investigation notes, and regulatory documents, reducing analyst turnaround time to same-day insights.
- Engineered embedding-based retrieval systems using vector databases (FAISS, Pinecone) to support scalable knowledge discovery and contextual decision-making workflows.
- Implemented advanced customer segmentation and behavioral modeling using PyTorch, autoencoders, and clustering techniques across 300K+ accounts, improving risk stratification and personalized engagement strategies.
- Architected scalable data pipelines on AWS (S3, Glue, Redshift, Airflow) for ingesting and transforming structured and semi-structured data, transitioning batch workflows to near real-time processing.
- Built and maintained feature stores and model-ready datasets, ensuring consistency across training and inference pipelines for 15+ ML use cases.
- Deployed production-grade ML systems using Docker, Kubernetes (EKS), and SageMaker, implementing model versioning, blue-green deployments, and automated rollback strategies.
- Established MLOps and AI observability frameworks, including model monitoring, drift detection, and performance tracking, to ensure reliability in regulated environments.
- Collaborated with cross-functional stakeholders (product, compliance, engineering) to implement model explainability (SHAP), governance controls, and audit-ready documentation, supporting model risk management and regulatory compliance.

LTIMindtree | Data Engineer (ML & Analytics)

May 2020 – Jul 2022 | India

- Designed and implemented AI-driven fraud detection systems using PyTorch, autoencoders, and Isolation Forest, identifying 8,200+ anomalous financial events per quarter and improving AML compliance and detection accuracy.
- Developed end-to-end NLP pipelines using Hugging Face Transformers and spaCy for classification, sentiment analysis, and summarization of customer complaints, reducing manual triage time from 4 hours to under 30 minutes and enhancing regulatory reporting accuracy.
- Built time-series forecasting models (LSTM, ARIMA, Prophet) to predict cash flow trends across 120+ corporate accounts, improving forecasting precision and supporting proactive liquidity risk management.
- Automated feature engineering and ML experimentation workflows using Databricks, SQL, and MLflow, reducing data preparation efforts by over 100 hours per month and improving model reproducibility.
- Implemented scalable MLOps pipelines using Docker, Kubernetes, GitHub Actions, and AWS SageMaker, enabling continuous integration, automated deployments, and model lifecycle management.
- Engineered high-performance ETL pipelines using Apache Spark and AWS Glue, processing 200K+ daily transactional records and delivering clean, structured datasets for downstream analytics and machine learning applications.
- Optimized database performance (PostgreSQL, MongoDB) through indexing, partitioning, and query tuning, reducing query latency from 120 ms to 40 ms and improving reporting efficiency.
- Developed automated data quality and validation frameworks using Python and SQL, ensuring data integrity, detecting anomalies early, and maintaining consistency across ML pipelines and reporting systems.

PROJECTS

Fraud Detection in Financial Transactions using AI/ML

- Formulated and implemented machine learning models using Python, PyTorch, and XGBoost to detect fraudulent transactions across 1M+ records, improving detection efficiency for high-risk activities.
- Devised feature engineering pipelines leveraging behavioral patterns, transaction sequences, and network-based features, enabling models to capture subtle anomalies in customer activity.

EDUCATION

Master of Science in Information Systems | Trine University, Angola, IN,

Dec 2023